# A Personalized Research Framework for Automated Prioritization of Urgent, Important, and Informational Emails

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## Abstract

Managing an overflowing inbox is a universal problem, mainly because "priority" is subjective to everyone's own opinion and hard to define. While AI could help in some aspects, progress has been stalled because there aren't enough datasets that are clearly labeled to teach models what matters. In this paper, we introduce a weakly supervised framework designed to spot urgent, informational and important emails within the Enron corpus. Rather than chasing a vague definition of priority, we break it down into two concrete parts: what needs to be done and when it's due. Using a Snorkel-inspired weak supervision pipeline, we generate high-quality labels automatically to overcome the data shortage, similar to recent work on email intent detection with weak supervision (Shu et al., 2020) We also look at the power of personalization. By comparing general models against those that factor in who is talking to whom, we quantify exactly how much "social context" helps an AI understand intent in messy, everyday professional text.

## 1 Introduction

In the modern workplace, professionals are constantly inundated with information, making it critical to distinguish between purely informational content and actionable requests, and to separately reason about perceived importance and urgency (Leonardi and Treem, 2022). However, "urgency" is rarely determined by text alone; it is a product of both linguistic signals (e.g., "by end of day") and the relational context between the sender and receiver.

To resolve this we propose a three stage strategy:

1. **Task Redefinition:** We shift from the subjective concept of "priority" to a concrete multi-label classification task focused on intent-based actionability and temporal deadlines.
2. **Weak Supervision:** To overcome the lack of labeled data, we use a programmatic labeling pipeline, inspired by recent weakly supervised email intent detection frameworks (Shu et al., 2020), to scan the Enron corpus for linguistic cues such as modal verbs and urgency lexicons to generate high-quality pseudo-labels.
3. **Personalization:** We investigate how integrating sender–receiver metadata and relational hierarchy improves model performance compared to traditional, text-only baselines, extending recent work that leverages user interaction signals for email intent detection (Shu et al., 2020).

## 2 Methodology

Our project aims to create a structured and reproducible framework for deriving weak labels for email actionability and temporal urgency, and to evaluate NLP models on these labels. The methodology consists of four major components: dataset preprocessing, weak supervision labeling functions, multi-label classifier training, and evaluation results.

### 2.1 Dataset and Preprocessing

We use the Enron Email Dataset as our primary corpus, following recent work that applies modern large language models for intent-based email categorization on Enron (Patel and Others, 2024). Since Enron emails are noisy, we apply several preprocessing steps, similar to recent NLP pipelines for Enron-based email classification and spam/intent modeling (Patel and Others, 2024; Chirra and Others, 2025):

- Removal email signatures, legal disclaimers, and forwarding headers.
- Thread deduplication to avoid repeated content inflating model performance.

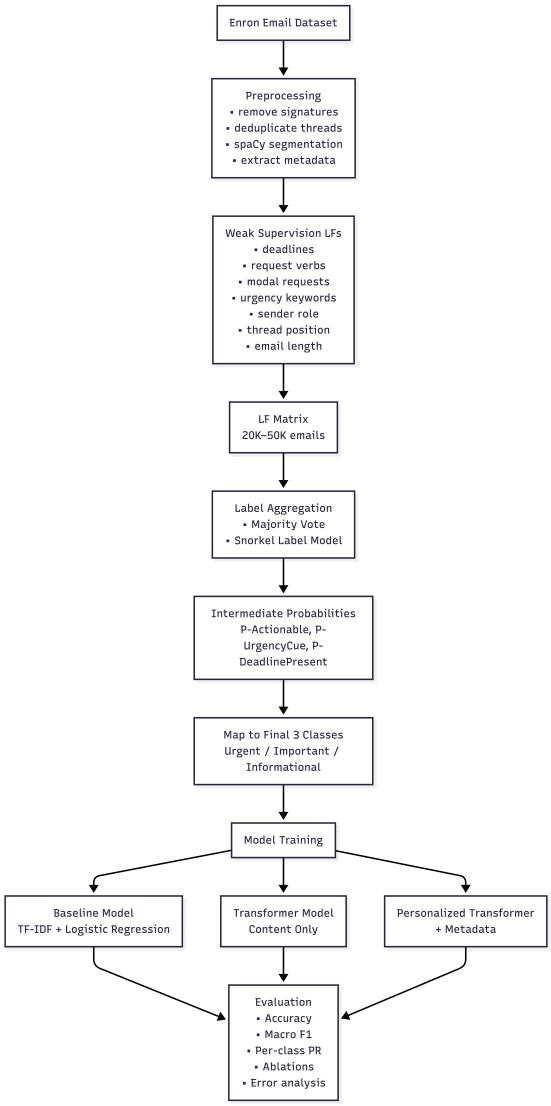


Figure 1: Overall project pipeline showing preprocessing, weak labeling, label aggregation, final 3-label mapping, model training, and evaluation.

- Sentence segmentation using spaCy for fine grained rule matching.
- Extraction of metadata including sender, recipients, subject line, and timestamps.

## 2.2 Weak Supervision for Intermediate Signals

Since the dataset contains no urgency labels, we use rule-based labeling functions (LFs) inspired by Snorkel to generate intermediate probabilistic annotations for:

- Actionability $\in \{0, 1\}$
- UrgencyCue $\in \{0, 1\}$
- DeadlinePresent $\in \{0, 1\}$

### 2.2.1 Labeling Functions (LFs)

LLMs are used only to generate candidate regex patterns, which are manually reviewed.

Each LF returns `1`, `0`. We design functions in the following categories:

**Temporal and Deadline Signals**

- **LF_deadline**: Detects explicit dates/times ("by Monday", "EOD", "2 PM tomorrow").

**Request-Based Signals**

- **LF_request_verb**: Matches verbs like *submit, approve, review, send.*
- **LF_modal_request**: Matches phrases such as "could you", "please", "I need you to".

**Urgency Keywords**

- **LF_urgency_keywords**: Detects "urgent", "ASAP", "immediately", etc.

**Metadata and Structural Cues**

- **LF_sender_role**: Emails from high-importance senders receive stronger actionability weights.
- **LF_thread_position**: Initial messages in a thread are more likely actionable.
- **LF_email_length**: Very short transactional emails often indicate action.

### 2.2.2 Label Aggregation

We apply all LFs to 20K–50K emails to produce an LF matrix. We aggregate LF outputs using majority vote (baseline) and Snorkel's generative label model (Shu et al., 2020), which has been shown to effectively denoise weak labels for email intent detection.

The label model produces probabilistic intermediate annotations: P(Actionable), P(UrgencyCue) and P(DeadlinePresent)

### 2.2.3 Gold Test Set

We manually annotate $\sim$300 emails for:

- actionable vs. non-actionable
- urgent vs. non-urgent
- presence of deadlines

This set is used only for evaluation.

### 2.3 Mapping Intermediate Signals to Final Labels

The final classification task is **three-way single-label classification:**

$$\text{Class} \in \{\text{Urgent}, \text{Important}, \text{Informational}\}.$$

We derive final weak labels using the intermediate signals:

- **Urgent**: high $P(\text{UrgencyCue})$ or $P(\text{DeadlinePresent})$.
- **Important**: high $P(\text{Actionable})$ but low urgency.
- **Informational**: low actionable and low urgency.

This mapping converts noisy binary signals into a stable 3-class target suitable for supervised learning.

### 2.4 Classification Models

We train three model variants to predict the final class.

#### 2.4.1 Baseline Model

A TF-IDF + Logistic Regression classifier provides a lower-bound reference.

#### 2.4.2 Transformer Model (Content Only)

We fine-tune a BERT / RoBERTa model using:

- email text
- intermediate weak labels mapped to the 3-way target

#### 2.4.3 Personalized Transformer Model

To test whether personalization improves classification, we extend the model with metadata features, building on evidence that incorporating user interactions and contextual signals improves email intent modeling (Shu et al., 2020).

- sender importance
- past reply frequency (synthetic)
- thread depth

### 2.5 Experiments and Evaluation

Our evaluation focuses on measuring how well each model predicts the final three classes (Urgent, Important, Informational) derived from our weak labels. We keep the evaluation simple and practical, aligned with the scope of a semester project.

#### 2.5.1 Train–Validation–Test Split

We use:

- 90% of weakly labeled data for training
- 10% of weakly labeled data for development testing
- the manually annotated 300 email gold set as the final test set

#### 2.5.2 Metrics

We report the following standard metrics:

- overall accuracy
- macro F1 score
- per-class precision and recall

These metrics help capture class imbalance and ensure each label is evaluated fairly.

#### 2.5.3 Models Compared

We compare three progressively stronger systems:

- **Baseline**: TF-IDF + Logistic Regression
- **Transformer (Content Only)**: BERT / RoBERTa fine-tuned on weak labels
- **Personalized Transformer**: BERT with additional metadata features

#### 2.5.4 Ablation Study (Simplified)

To understand which signals matter most, we run two small ablations:

**Effect of Temporal Cues** Train models:

- with deadline/urgency LFs included
- without temporal LFs

**Effect of Personalization** Compare:

- content-only model
- content + sender metadata

This small-scale inspection helps explain model weaknesses and guides future improvements.

## References


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